

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – TECHNICAL PRESENTATION

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 3 – Technical Presentation
Weightage:	10% of CIA	Total Marks:	100 (1 Question × 100 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	K Abishek	Reg. No.:	192521120
Section / Batch:	IT	Date:	7/23/2026

Instructions to Students

- Answer the given question through a technical presentation. Total Marks: 100.
- Clearly define the problem and explain the concept of algorithm growth functions.
- Present comparative analysis using appropriate representations such as graphs, tables, or charts.
- Use suitable tools (PowerPoint, visualization tools, or software) to support your explanation.
- Ensure technical accuracy, logical flow, and proper interpretation of results.
- Maintain clarity in communication, proper slide organization, and professional delivery.
- Time management, confidence, and audience engagement will be considered during evaluation.

Question Type	Focus Area	Questions	Marks
Technical Presentation	Analyze and compare growth functions using mathematical techniques and asymptotic concepts and with graphical and visual interpretation	Q1	100
GRAND TOTAL:		100 Marks	

SECTION A – Technical Presentation

Q1. Visualization-Based Technical Presentation on Growth Functions

Analyze growth functions $O(\log n)$, $O(n)$, $O(n \log n)$, $O(n^2)$, and $O(2^n)$ using visualization tools in a technical presentation format. Clearly define the purpose of comparing these functions for understanding algorithm efficiency. Use graphs, tables, or charts to show how each function behaves as the input size increases. Explain which functions remain practical for large-scale problems and justify your reasoning. Present the conclusions clearly and professionally by linking growth behavior to algorithm selection.

Code of Conduct

I K Abishek/192521120 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K Abishek

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Marks Evaluation:

Question No.	Technical Content Accuracy (30%)	Problem Identification (25%)	Use of Tools (20%)	Communication (15%)	Professionalism (10%)	Total Marks (100)
Q1	4	4	4	2	3	90
Overall Total (100)						90

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____



DIVIDE AND CONQUER COMPARISON WITH DIFFERENT RECURRENCE STRUCTURES

CSA0620 - DESIGN AND ANALYSIS OF ALGORITHM

Presented by

K Abishek (192521120)



Index



- ◆ Introduction
- ◆ Recurrence Structures
- ◆ Recursion Tree Comparison
- ◆ Master Theorem and Efficiency Comparison
- ◆ Conclusion
- ◆ References

Introduction

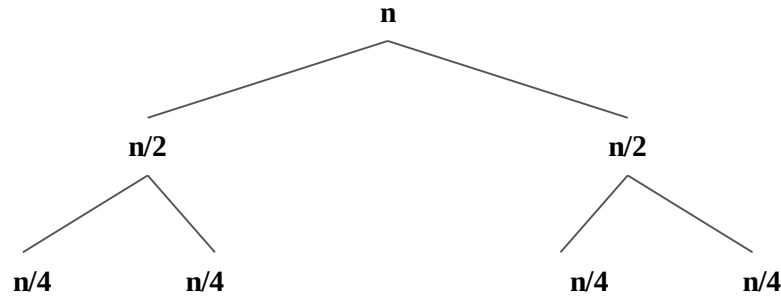
- ◆ Divide-and-conquer algorithms solve a problem by breaking it into smaller subproblems of the same type, solving each recursively, and combining the results.
- ◆ Their running time is written as a recurrence relation: $T(n) = a \cdot T(n/b) + f(n)$.
- ◆ a = number of subproblems created at each call.
- ◆ b = factor by which the problem size shrinks.
- ◆ $f(n)$ = cost of dividing the problem and combining the results.

Recurrence Structures Defined

- ◆ Algorithm A: $T(n) = 2T(n/2) + n$ — splits into 2 subproblems, each of size $n/2$ (example: Merge Sort).
- ◆ Algorithm B: $T(n) = 4T(n/2) + n$ — splits into 4 subproblems, each of size $n/2$ (example: naive matrix multiplication style split).
- ◆ Both algorithms shrink the problem by the same factor ($b = 2$) and have the same combine cost, $f(n) = n$.
- ◆ The only difference is the number of subproblems (a): 2 versus 4.

Recursion Tree Comparison

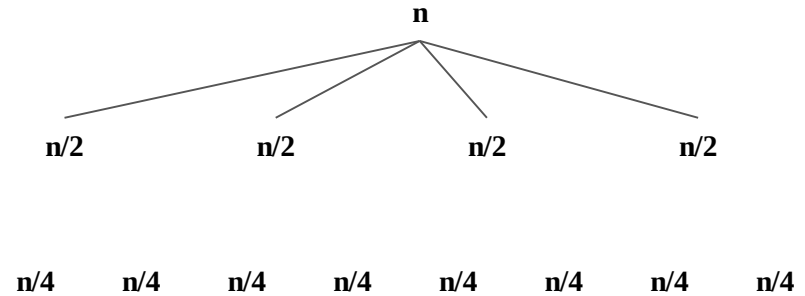
$$T(n) = 2T(n/2) + n$$



Level cost stays about n at every level (constant per level).

$$\log n \text{ levels} \times O(n) = O(n \log n)$$

$$T(n) = 4T(n/2) + n$$



Level cost quadruples each level (grows fast).

$$\text{Cost dominated by leaves} = O(n^2)$$

Master Theorem and Efficiency Comparison



- ◆ Master Theorem: for $T(n) = aT(n/b) + f(n)$, compare $f(n)$ with $n^{\log_m a}$.
- ◆ Algorithm A ($a=2, b=2$): $\log_2 2 = 1$, and $f(n) = n = \Theta(n^1) \rightarrow \text{Case 2} \rightarrow T(n) = \Theta(n \log n)$.
- ◆ Algorithm B ($a=4, b=2$): $\log_2 4 = 2$, and $f(n) = n = O(n^{2-\epsilon}) \rightarrow \text{Case 1} \rightarrow T(n) = \Theta(n^2)$.
- ◆ Conclusion: Algorithm A, $T(n) = 2T(n/2) + n$, is more efficient than Algorithm B..

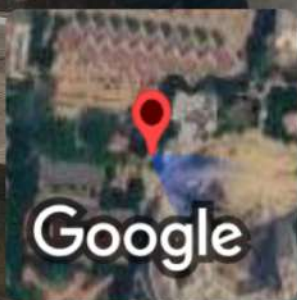
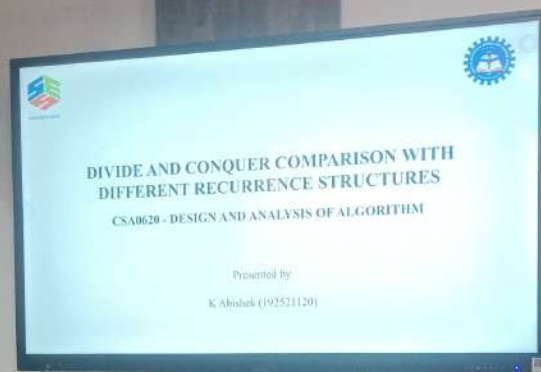
Result

- ◆ Algorithm A ($2T(n/2)+n$) runs in $\Theta(n \log n)$; Algorithm B ($4T(n/2)+n$) runs in $\Theta(n^2)$
- ◆ Fewer subproblems per call ($a=2$ vs $a=4$) keeps recursion tree cost balanced across levels instead of exploding at the leaves
- ◆ Algorithm A is more efficient overall, especially as n grows large

References

- ◆ Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2009). Introduction to Algorithms (3rd ed.). MIT Press.
- ◆ Sedgewick, R., & Wayne, K. (2011). Algorithms (4th ed.). Addison-Wesley.
- ◆ Kleinberg, J., & Tardos, E. (2005). Algorithm Design. Pearson Education.
- ◆ Levitin, A. (2012). Introduction to the Design and Analysis of Algorithms (3rd ed.). Pearson.
- ◆ GeeksforGeeks. Master Theorem for Time Complexity Analysis. <https://www.geeksforgeeks.org>

Thank You



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